The quantities Γ and τ are called the ***reflection*** and ***transmission coefficients***. For lossless dielectric media, η_1 and η_2 are real; consequently, both Γ and τ are real also. As we see in Section 8-1.4, the expressions given by Eqs. (8.8a) and (8.8b) are equally applicable when the media are conductive, even though in that case η_1 and η_2 may be complex. Hence, Γ and τ may be complex as well. From Eqs. (8.8a) and (8.8b), it is easily shown that Γ and τ are interrelated as

$$\tau = 1 + \Gamma \qquad \textbf{(normal incidence)}. \tag{8.9}$$

For nonmagnetic media,

$$\eta_1 = \frac{\eta_0}{\sqrt{\varepsilon_{r_1}}}, \quad \text{and} \quad \eta_2 = \frac{\eta_0}{\sqrt{\varepsilon_{r_2}}},$$

where η_0 is the intrinsic impedance of free space. In this case, Eq. (8.8a) may be expressed as

$$\Gamma = \frac{\sqrt{\varepsilon_{r_1}} - \sqrt{\varepsilon_{r_2}}}{\sqrt{\varepsilon_{r_1}} + \sqrt{\varepsilon_{r_2}}} \qquad \textbf{(nonmagnetic media)}. \tag{8.10}$$

8-1.2 Transmission-Line Analogue

The transmission-line configuration shown in **Fig. 8-4(b)** consists of a lossless transmission line with characteristic impedance Z_{01} connected at $z = 0$ to an infinitely long lossless transmission line with characteristic impedance Z_{02}. The input impedance of an infinitely long line is equal to its characteristic impedance. Hence, at $z = 0$, the voltage reflection coefficient (looking toward the boundary from the vantage point of the first line) is

$$\Gamma = \frac{Z_{02} - Z_{01}}{Z_{02} + Z_{01}},$$

which is identical in form to Eq. (8.8a). The analogy between plane waves and waves on transmission lines does not end here. To demonstrate the analogy further, equations pertinent to the two cases are summarized in **Table 8-1**. Comparison of the two columns shows that there is a one-to-one correspondence between the transmission-line quantities $(\widetilde{V}, \widetilde{I}, \beta, Z_0)$ and the plane-wave quantities $(\widetilde{E}, \widetilde{H}, k, \eta)$.

► This correspondence allows us to use the techniques developed in Chapter 2, including the Smith-chart method for calculating impedance transformations, to solve plane-wave propagation problems. ◄

The simultaneous presence of incident *and* reflected waves in medium 1 (**Fig. 8-4(a)**) gives rise to a standing-wave pattern. By analogy with the transmission-line case, the ***standing-wave ratio*** in medium 1 is defined as

$$S = \frac{|\widetilde{E}_1|_{\max}}{|\widetilde{E}_1|_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|}. \tag{8.15}$$

► If the two media have equal impedances ($\eta_1 = \eta_2$), then $\Gamma = 0$ and $S = 1$, and if medium 2 is a perfect conductor with $\eta_2 = 0$ (which is equivalent to a short-circuited transmission line), then $\Gamma = -1$ and $S = \infty$. ◄

The distance from the boundary to where the magnitude of the electric field intensity in medium 1 is a maximum (denoted $l_{\max}$) is described by the same expression as that given by Eq. (2.70) for the voltage maxima on a transmission line, namely

$$-z = l_{\max} = \frac{\theta_r + 2n\pi}{2k_1} = \frac{\theta_r \lambda_1}{4\pi} + \frac{n\lambda_1}{2}, \qquad \begin{cases} n = 1, 2, \ldots & \text{if } \theta_r < 0, \\ n = 0, 1, 2, \ldots & \text{if } \theta_r \geq 0, \end{cases} \tag{8.16}$$

where $\lambda_1 = 2\pi/k_1$ and θ_r is the phase angle of Γ (i.e., $\Gamma = |\Gamma| e^{j\theta_r}$, and θ_r is bounded in the range $-\pi < \theta_r \leq \pi$). The expression for $l_{\max}$ is valid not only when the two media are lossless dielectrics, but also when medium 1 is a low-loss dielectric. Moreover, medium 2 may be either a dielectric or a conductor. When both media are lossless dielectrics, $\theta_r = 0$ if $\eta_2 > \eta_1$, and $\theta_r = \pi$ if $\eta_2 < \eta_1$.

The spacing between adjacent maxima is $\lambda_1/2$, and the spacing between a maximum and the nearest minimum is $\lambda_1/4$. The electric-field minima occur at

$$l_{\min} = \begin{cases} l_{\max} + \lambda_1/4, & \text{if } l_{\max} < \lambda_1/4, \\ l_{\max} - \lambda_1/4, & \text{if } l_{\max} \geq \lambda_1/4. \end{cases} \tag{8.17}$$

8-1.3 Power Flow in Lossless Media

Medium 1 in **Fig. 8-4(a)** is host to the incident and reflected waves, which together comprise the total electric and magnetic fields $\widetilde{\mathbf{E}}_1(z)$ and $\widetilde{\mathbf{H}}_1(z)$ given by Eqs. (8.11a) and (8.12a) of **Table 8-1**. Using Eq. (7.100), the net average power density

量 Γ 和 τ 称为反射系数和透射系数。对于无损电介质，η_1和η_2是实数；因此，Γ 和 τ 也是实数。正如我们在第 8-1.4 节中看到的，方程给出的表达式。当介质导电时，（8.8a）和（8.8b）同样适用，即使在这种情况下η_1和η_2可能很复杂。因此，Γ 和 τ 也可能很复杂。从方程式。（8.8a）和（8.8b），很容易证明Γ和τ是相互关联的

$$\tau = 1 + \Gamma \qquad \textbf{(normal incidence)}. \tag{8.9}$$

对于非磁性介质，

$$\eta_1 = \frac{\eta_0}{\sqrt{\varepsilon_{r_1}}}, \quad \text{and} \quad \eta_2 = \frac{\eta_0}{\sqrt{\varepsilon_{r_2}}},$$

其中 η_0 是自由空间的固有阻抗。在这种情况下，等式。(8.8a) 可表示为

$$\Gamma = \frac{\sqrt{\varepsilon_{r_1}} - \sqrt{\varepsilon_{r_2}}}{\sqrt{\varepsilon_{r_1}} + \sqrt{\varepsilon_{r_2}}} \qquad \textbf{(nonmagnetic media)}. \tag{8.10}$$

8-1.2 传输线模拟

图 8-4(b) 所示的传输线配置由一条特征阻抗为 Z_{01} 的无损传输线在 $z = 0$ 处连接到一条特征阻抗为 Z_{02} 的无限长无损传输线组成。无限长线路的输入阻抗等于其特性阻抗。因此，在 $z = 0$ 处，电压反射系数（从第一条线的有利位置看向边界）为

$$\Gamma = \frac{Z_{02} - Z_{01}}{Z_{02} + Z_{01}},$$

其形式与等式相同。(8.8a)。平面波和传输线上的波之间的类比还没有结束。为了进一步证明这种类比，表 8-1 总结了与这两种情况相关的方程。两列的比较表明，传输线量（$\widetilde{V}$、$\widetilde{I}$、β、Z_0）和平面波量（$\widetilde{E}$、$\widetilde{H}$、k、η）之间存在一一对应关系。

► 这种对应关系使我们能够使用第 2 章中开发的技术（包括用于计算阻抗变换的史密斯圆图方法）来解决平面波传播问题。 ◄

介质 1 中同时存在入射波和反射波（图 8-4(a)），从而产生驻波图案。

与传输线情况类比，介质 1 中的驻波比定义为

$$S = \frac{|\widetilde{E}_1|_{\max}}{|\widetilde{E}_1|_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|}. \tag{8.15}$$

► 如果两种介质具有相等的阻抗（$\eta_1 = \eta_2$），则 $\Gamma = 0$且$S = 1$，并且如果介质2是理想导体且$\eta_2 = 0$（相当于短路传输线），则$\Gamma = -1$和$S = \infty$。 ◄

从边界到介质 1 中电场强度的最大值（表示为 $l_{\max}$）的距离可以用与等式 1 给出的相同的表达式来描述。(2.70) 为传输线上的电压最大值，即

$$-z = l_{\max} = \frac{\theta_r + 2n\pi}{2k_1} = \frac{\theta_r \lambda_1}{4\pi} + \frac{n\lambda_1}{2}, \qquad \begin{cases} n = 1, 2, \ldots & \text{if } \theta_r < 0, \\ n = 0, 1, 2, \ldots & \text{if } \theta_r \geq 0, \end{cases} \tag{8.16}$$

其中，$\lambda_1 = 2\pi/k_1$ 和 θ_r 是 Γ（的相位角，即 $\Gamma = |\Gamma| e^{j\theta_r}$，而 θ_r 的范围为 $-\pi < \theta_r \leq \pi$)。$l_{\max}$ 的表达式不仅在两种介质是无损电介质时有效，而且在介质 1 是低损耗电介质时也有效。此外，介质2可以是电介质或导体。当两种介质都是无损电介质时，如果 $\eta_2 > \eta_1$，则 $\theta_r = 0$；如果 $\eta_2 < \eta_1$，则 $\theta_r = \pi$。

相邻最大值之间的间距为$\lambda_1/2$，最大值与最近的最小值之间的间距为$\lambda_1/4$。电场最小值出现在

$$l_{\min} = \begin{cases} l_{\max} + \lambda_1/4, & \text{if } l_{\max} < \lambda_1/4, \\ l_{\max} - \lambda_1/4, & \text{if } l_{\max} \geq \lambda_1/4. \end{cases} \tag{8.17}$$

8-1.3 无损媒体中的功率流

图 8-4(a) 中的介质 1 是入射波和反射波的宿主，它们共同构成了方程 1 给出的总电场和磁场 $\mathbf{E}_1(z)$ 和 $\mathbf{H}_1(z)$。表 8-1的(8.11a)和(8.12a)。使用方程式(7.100)，净平均功率密度

Table 8-1 Analogy between plane-wave equations for normal incidence and transmission-line equations with both under lossless conditions.

Plane Wave (Fig. 8-4(a))		Transmission Line (Fig. 8-4(b))	
$\widetilde{\mathbf{E}}_1(z) = \hat{\mathbf{x}} E_0^{\mathrm{i}}(e^{-jk_1z} + \Gamma e^{jk_1z})$	(8.11a)	$\widetilde{V}_1(z) = V_0^+(e^{-j\beta_1z} + \Gamma e^{j\beta_1z})$	(8.11b)
$\widetilde{\mathbf{H}}_1(z) = \hat{\mathbf{y}} \dfrac{E_0^{\mathrm{i}}}{\eta_1}(e^{-jk_1z} - \Gamma e^{jk_1z})$	(8.12a)	$\widetilde{I}_1(z) = \dfrac{V_0^+}{Z_{01}}(e^{-j\beta_1z} - \Gamma e^{j\beta_1z})$	(8.12b)
$\widetilde{\mathbf{E}}_2(z) = \hat{\mathbf{x}} \tau E_0^{\mathrm{i}} e^{-jk_2z}$	(8.13a)	$\widetilde{V}_2(z) = \tau V_0^+ e^{-j\beta_2z}$	(8.13b)
$\widetilde{\mathbf{H}}_2(z) = \hat{\mathbf{y}} \tau \dfrac{E_0^{\mathrm{i}}}{\eta_2} e^{-jk_2z}$	(8.14a)	$\widetilde{I}_2(z) = \tau \dfrac{V_0^+}{Z_{02}} e^{-j\beta_2z}$	(8.14b)
$\Gamma = (\eta_2 - \eta_1)/(\eta_2 + \eta_1)$		$\Gamma = (Z_{02} - Z_{01})/(Z_{02} + Z_{01})$	
$\tau = 1 + \Gamma$		$\tau = 1 + \Gamma$	
$k_1 = \omega\sqrt{\mu_1\varepsilon_1}$, $\quad k_2 = \omega\sqrt{\mu_2\varepsilon_2}$		$\beta_1 = \omega\sqrt{\mu_1\varepsilon_1}$, $\quad \beta_2 = \omega\sqrt{\mu_2\varepsilon_2}$	
$\eta_1 = \sqrt{\mu_1/\varepsilon_1}$, $\quad \eta_2 = \sqrt{\mu_2/\varepsilon_2}$		Z_{01} and Z_{02} depend on transmission-line parameters	

flowing in medium 1 is

$$\mathbf{S}_{\mathrm{av}_1}(z) = \tfrac{1}{2}\mathfrak{Re}[\widetilde{\mathbf{E}}_1(z) \times \widetilde{\mathbf{H}}_1^*(z)]$$
$$= \tfrac{1}{2}\mathfrak{Re}\left[\hat{\mathbf{x}} E_0^{\mathrm{i}}(e^{-jk_1z} + \Gamma e^{jk_1z})\hat{\mathbf{y}} \frac{E_0^{\mathrm{i}*}}{\eta_1}(e^{jk_1z} - \Gamma^* e^{-jk_1z})\right]$$
$$= \hat{\mathbf{z}} \frac{|E_0^{\mathrm{i}}|^2}{2\eta_1}(1 - |\Gamma|^2), \tag{8.18}$$

which is analogous to Eq. (2.106) for the lossless transmission-line case. The first and second terms inside the bracket in Eq. (8.18) represent the average power density of the incident and reflected waves, respectively. Thus,

$$\mathbf{S}_{\mathrm{av}_1} = \mathbf{S}_{\mathrm{av}}^{\mathrm{i}} + \mathbf{S}_{\mathrm{av}}^{\mathrm{r}} \tag{8.19a}$$

with

$$\mathbf{S}_{\mathrm{av}}^{\mathrm{i}} = \hat{\mathbf{z}} \frac{|E_0^{\mathrm{i}}|^2}{2\eta_1}, \tag{8.19b}$$

$$\mathbf{S}_{\mathrm{av}}^{\mathrm{r}} = -\hat{\mathbf{z}}|\Gamma|^2 \frac{|E_0^{\mathrm{i}}|^2}{2\eta_1} = -|\Gamma|^2 \mathbf{S}_{\mathrm{av}}^{\mathrm{i}}. \tag{8.19c}$$

Even though Γ is purely real when both media are lossless dielectrics, we chose to treat it as complex, thereby providing in Eq. (8.19c) an expression that is also valid when medium 2 is conducting.

The average power density of the transmitted wave in medium 2 is

$$\mathbf{S}_{\mathrm{av}_2}(z) = \tfrac{1}{2}\mathfrak{Re}[\widetilde{\mathbf{E}}_2(z) \times \widetilde{\mathbf{H}}_2^*(z)]$$
$$= \tfrac{1}{2}\mathfrak{Re}\left[\hat{\mathbf{x}} \tau E_0^{\mathrm{i}} e^{-jk_2z} \times \hat{\mathbf{y}} \tau^* \frac{E_0^{\mathrm{i}*}}{\eta_2} e^{jk_2z}\right] = \hat{\mathbf{z}}|\tau|^2 \frac{|E_0^{\mathrm{i}}|^2}{2\eta_2}. \tag{8.20}$$

Through the use of Eqs. (8.8a and b), it can be easily shown that for lossless media

$$\frac{\tau^2}{\eta_2} = \frac{1 - \Gamma^2}{\eta_1}, \qquad \textbf{(lossless media)} \tag{8.21}$$

which leads to

$$\mathbf{S}_{\mathrm{av}_1} = \mathbf{S}_{\mathrm{av}_2}.$$

This result is expected from considerations of power conservation.

Example 8-1: Radar Radome Design

A 10 GHz aircraft radar uses a narrow-beam scanning antenna mounted on a gimbal behind a dielectric radome, as shown in Fig. 8-5. Even though the radome shape is far from planar, it is approximately planar over the narrow extent of the radar beam. If the radome material is a lossless dielectric with $\varepsilon_r = 9$ and

表 8-1 法向入射平面波方程与无损条件下的传输线方程之间的类比。

Plane Wave (Fig. 8-4(a))		Transmission Line (Fig. 8-4(b))	
$\widetilde{\mathbf{E}}_1(z) = \hat{\mathbf{x}} E_0^{\mathrm{i}}(e^{-jk_1z} + \Gamma e^{jk_1z})$	(8.11a)	$\widetilde{V}_1(z) = V_0^+(e^{-j\beta_1z} + \Gamma e^{j\beta_1z})$	(8.11b)
$\widetilde{\mathbf{H}}_1(z) = \hat{\mathbf{y}} \dfrac{E_0^{\mathrm{i}}}{\eta_1}(e^{-jk_1z} - \Gamma e^{jk_1z})$	(8.12a)	$\widetilde{I}_1(z) = \dfrac{V_0^+}{Z_{01}}(e^{-j\beta_1z} - \Gamma e^{j\beta_1z})$	(8.12b)
$\widetilde{\mathbf{E}}_2(z) = \hat{\mathbf{x}} \tau E_0^{\mathrm{i}} e^{-jk_2z}$	(8.13a)	$\widetilde{V}_2(z) = \tau V_0^+ e^{-j\beta_2z}$	(8.13b)
$\widetilde{\mathbf{H}}_2(z) = \hat{\mathbf{y}} \tau \dfrac{E_0^{\mathrm{i}}}{\eta_2} e^{-jk_2z}$	(8.14a)	$\widetilde{I}_2(z) = \tau \dfrac{V_0^+}{Z_{02}} e^{-j\beta_2z}$	(8.14b)
$\Gamma = (\eta_2 - \eta_1)/(\eta_2 + \eta_1)$		$\Gamma = (Z_{02} - Z_{01})/(Z_{02} + Z_{01})$	
$\tau = 1 + \Gamma$		$\tau = 1 + \Gamma$	
$k_1 = \omega\sqrt{\mu_1\varepsilon_1}$, $\quad k_2 = \omega\sqrt{\mu_2\varepsilon_2}$		$\beta_1 = \omega\sqrt{\mu_1\varepsilon_1}$, $\quad \beta_2 = \omega\sqrt{\mu_2\varepsilon_2}$	
$\eta_1 = \sqrt{\mu_1/\varepsilon_1}$, $\quad \eta_2 = \sqrt{\mu_2/\varepsilon_2}$		Z_{01} and Z_{02} depend on transmission-line parameters	

在介质 1 中流动的是

$$\mathbf{S}_{\mathrm{av}_1}(z) = \tfrac{1}{2}\mathfrak{Re}[\widetilde{\mathbf{E}}_1(z) \times \widetilde{\mathbf{H}}_1^*(z)]$$
$$= \tfrac{1}{2}\mathfrak{Re}\left[\hat{\mathbf{x}} E_0^{\mathrm{i}}(e^{-jk_1z} + \Gamma e^{jk_1z})\hat{\mathbf{y}} \frac{E_0^{\mathrm{i}*}}{\eta_1}(e^{jk_1z} - \Gamma^* e^{-jk_1z})\right]$$
$$= \hat{\mathbf{z}} \frac{|E_0^{\mathrm{i}}|^2}{2\eta_1}(1 - |\Gamma|^2), \tag{8.18}$$

这类似于等式。 (2.106) 适用于无损传输线情况。方程中括号内的第一项和第二项(8.18) 分别表示入射波和反射波的平均功率密度。因此，

$$\mathbf{S}_{\mathrm{av}_1} = \mathbf{S}_{\mathrm{av}}^{\mathrm{i}} + \mathbf{S}_{\mathrm{av}}^{\mathrm{r}} \tag{8.19a}$$

和

$$\mathbf{S}_{\mathrm{av}}^{\mathrm{i}} = \hat{\mathbf{z}} \frac{|E_0^{\mathrm{i}}|^2}{2\eta_1}, \tag{8.19b}$$

$$\mathbf{S}_{\mathrm{av}}^{\mathrm{r}} = -\hat{\mathbf{z}}|\Gamma|^2 \frac{|E_0^{\mathrm{i}}|^2}{2\eta_1} = -|\Gamma|^2 \mathbf{S}_{\mathrm{av}}^{\mathrm{i}}. \tag{8.19c}$$

尽管当两种介质都是无损电介质时 Γ 是纯实数，但我们选择将其视为复数，从而在等式中提供： (8.19c) 当介质 2 导通时该表达式也有效。

发射波的平均功率密度

中 2 是

$$\mathbf{S}_{\mathrm{av}_2}(z) = \tfrac{1}{2}\mathfrak{Re}[\widetilde{\mathbf{E}}_2(z) \times \widetilde{\mathbf{H}}_2^*(z)]$$
$$= \tfrac{1}{2}\mathfrak{Re}\left[\hat{\mathbf{x}} \tau E_0^{\mathrm{i}} e^{-jk_2z} \times \hat{\mathbf{y}} \tau^* \frac{E_0^{\mathrm{i}*}}{\eta_2} e^{jk_2z}\right] = \hat{\mathbf{z}}|\tau|^2 \frac{|E_0^{\mathrm{i}}|^2}{2\eta_2}. \tag{8.20}$$

通过使用方程式。 （8.8a和b），可以很容易地证明对于无损媒体

$$\frac{\tau^2}{\eta_2} = \frac{1 - \Gamma^2}{\eta_1}, \qquad \textbf{(lossless media)} \tag{8.21}$$

这导致

$$\mathbf{S}_{\mathrm{av}_1} = \mathbf{S}_{\mathrm{av}_2}.$$

这个结果是出于节能考虑的预期。

例 8-1：雷达天线罩设计

10GHz飞机雷达采用窄波束扫描天线，安装在介质天线罩后面的万向架上，如图8-5所示。尽管天线罩形状远非平面，但在雷达波束的狭窄范围内它近似平面。如果天线罩材料是无损电介质，$\varepsilon_r = 9$ 且

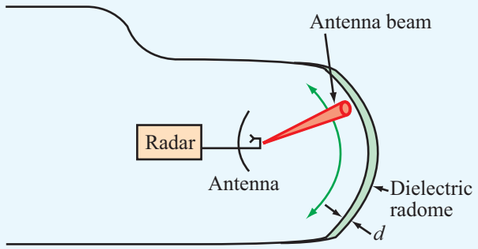


Figure 8-5 Antenna beam "looking" through an aircraft radome of thickness d (Example 8-1).

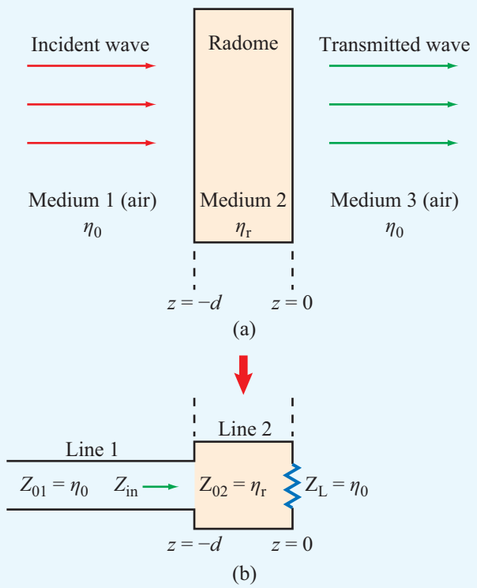


Figure 8-6 (a) Planar section of the radome of Fig. 8-5 at an expanded scale and (b) its transmission-line equivalent model (Example 8-1).

$\mu_r = 1$, choose its thickness d such that the radome appears transparent to the radar beam. Structural integrity requires d to be greater than 2.3 cm.

Solution: Figure 8-6(a) shows a small section of the radome on an expanded scale. The incident wave can be approximated as a plane wave propagating in medium 1 (air) with intrinsic impedance η_0. Medium 2 (the radome) is of thickness d and intrinsic impedance η_r, and medium 3 (air) is semi-infinite with intrinsic impedance η_0. Figure 8-6(b) shows an equivalent transmission-line model with $z = 0$ selected to coincide with the outside surface of the radome, and the load impedance $Z_L = \eta_0$ represents the input impedance of the semi-infinite air medium to the right of the radome.

For the radome to "appear" transparent to the incident wave, the reflection coefficient must be zero at $z = -d$, thereby guaranteeing total transmission of the incident power into medium 3. Since $Z_L = \eta_0$ in Fig. 8-6(b), no reflection takes place at $z = -d$ if $Z_{in} = \eta_0$, which can be realized by choosing $d = n\lambda_2/2$ [see Section 2-8.4], where λ_2 is the wavelength in medium 2 and n is a positive integer. At 10 GHz, the wavelength in air is $\lambda_0 = c/f = 3$ cm, while in the radome material it is

$$\lambda_2 = \frac{\lambda_0}{\sqrt{\varepsilon_r}} = \frac{3 \text{ cm}}{3} = 1 \text{ cm}.$$

Hence, by choosing $d = 5\lambda_2/2 = 2.5$ cm, the radome is both nonreflecting and structurally stable.

Example 8-2: Yellow Light Incident upon a Glass Surface

A beam of yellow light with wavelength 0.6 μm is normally incident in air upon a glass surface. Assume the glass is sufficiently thick as to ignore its back surface. If the surface is situated in the plane $z = 0$ and the relative permittivity of glass is 2.25, determine:

(a) the locations of the electric field maxima in medium 1 (air),
(b) the standing-wave ratio, and
(c) the fraction of the incident power transmitted into the glass medium.

Solution: (a) We begin by determining the values of η_1, η_2, and Γ:

$$\eta_1 = \sqrt{\frac{\mu_1}{\varepsilon_1}} = \sqrt{\frac{\mu_0}{\varepsilon_0}} \approx 120\pi \ (\Omega),$$

$$\eta_2 = \sqrt{\frac{\mu_2}{\varepsilon_2}} = \sqrt{\frac{\mu_0}{\varepsilon_0}} \cdot \frac{1}{\sqrt{\varepsilon_r}} \approx \frac{120\pi}{\sqrt{2.25}} = 80\pi \ (\Omega),$$

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{80\pi - 120\pi}{80\pi + 120\pi} = -0.2.$$

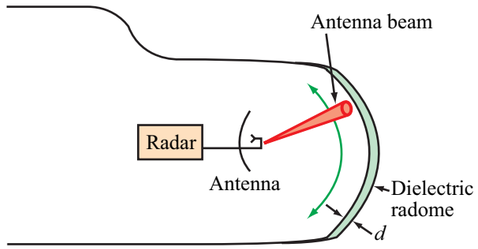


图 8-5 天线波束"看"过厚度为 d 的飞机天线罩（示例 8-1）。

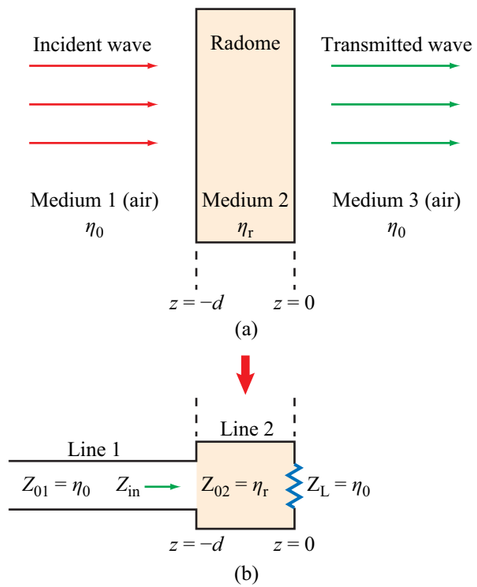


图 8-6 (a) 图 8-5 天线罩放大后的平面剖面，(b) 其传输线等效模型（例 8-1）。

$\mu_r = 1$，选择其厚度 d，使天线罩对雷达波束显得透明。结构完整性要求 d 大于 2.3 厘米。

解：图8-6(a)显示了天线罩的一小部分在扩大的规模上。入射波可以近似为在介质1（空气）中传播的具有固有阻抗η_0的平面波。介质 2（天线罩）的厚度为 d，固有阻抗为 η_r，介质 3（空气）为半无限，固有阻抗为 η_0。图 8-6(b) 显示了一个等效传输线模型，其中 z = 0 选择与天线罩外表面重合，负载阻抗 $Z_L = \eta_0$ 表示天线罩右侧半无限空气介质的输入阻抗。

为了让天线罩对入射波"显得"透明，在 $z = -d$ 处的反射系数必须为零，从而保证入射功率全部传输到介质 3 中。由于图 8-6（b）中的 $Z_L = \eta_0$，如果 $Z_{in} = \eta_0$，则在 $z = -d$ 处不会发生反射，这可以通过选择 $d = n\lambda_2/2$ 来实现2-8.4]，其中λ_2是介质2中的波长，n是正整数。在 10 GHz 时，空气中的波长为 $\lambda_0 = c/f = 3$ cm，而在天线罩材料中的波长为

$$\lambda_2 = \frac{\lambda_0}{\sqrt{\varepsilon_r}} = \frac{3 \text{ cm}}{3} = 1 \text{ cm}.$$

因此，通过选择 $d = 5\lambda_2/2 = 2.5$ cm，天线罩既不反射又结构稳定。

示例 8-2： 黄光入射到玻璃表面

波长为 0.6 μm 的黄光垂直在空气中入射到玻璃表面上。假设玻璃足够厚，可以忽略其背面。如果表面位于平面 z = 0 且玻璃的相对介电常数为 2.25，则确定：

(a) 介质 1（空气）中电场最大值的位置，(b) 驻波比，以及 (c) 传输到玻璃介质中的入射功率的比例。

解： (a) 我们首先确定 η_1、η_2 和 Γ 的值：

$$\eta_1 = \sqrt{\frac{\mu_1}{\varepsilon_1}} = \sqrt{\frac{\mu_0}{\varepsilon_0}} \approx 120\pi \ (\Omega),$$

$$\eta_2 = \sqrt{\frac{\mu_2}{\varepsilon_2}} = \sqrt{\frac{\mu_0}{\varepsilon_0}} \cdot \frac{1}{\sqrt{\varepsilon_r}} \approx \frac{120\pi}{\sqrt{2.25}} = 80\pi \ (\Omega),$$

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{80\pi - 120\pi}{80\pi + 120\pi} = -0.2.$$

Hence, $|\Gamma| = 0.2$ and $\theta_r = \pi$. From Eq. (8.16), the electric-field magnitude is maximum at

$$l_{\max} = \frac{\theta_r \lambda_1}{4\pi} + n\frac{\lambda_1}{2} = \frac{\lambda_1}{4} + n\frac{\lambda_1}{2} \qquad (n = 0, 1, 2, \ldots)$$

with $\lambda_1 = 0.6\ \mu$m.

(b)

$$S = \frac{1+|\Gamma|}{1-|\Gamma|} = \frac{1+0.2}{1-0.2} = 1.5.$$

(c) The fraction of the incident power transmitted into the glass medium is equal to the ratio of the transmitted power density given by Eq. (8.20) to the incident power density, $S^i_{av} = |E^i_0|^2/2\eta_1$:

$$\frac{S_{av_2}}{S^i_{av}} = \tau^2 \frac{|E^i_0|^2}{2\eta_2} \Big/ \left[\frac{|E^i_0|^2}{2\eta_1}\right] = \tau^2 \frac{\eta_1}{\eta_2}.$$

In view of Eq. (8.21),

$$\frac{S_{av_2}}{S^i_{av}} = 1 - |\Gamma|^2 = 1 - (0.2)^2 = 0.96, \text{ or } 96\%.$$

8-1.4 Boundary between Lossy Media

In Section 8-1.1, we considered a plane wave in a lossless medium incident normally on a planar boundary of another lossless medium. We now generalize our expressions to lossy media. In a medium with constitutive parameters $(\varepsilon, \mu, \sigma)$, the propagation constant $\gamma = \alpha + j\beta$ and the intrinsic impedance η_c are both complex. General expressions for α, β, and η_c are given by Eqs. (7.66a), (7.66b), and (7.70), respectively, and approximate expressions are given in **Table 7-2** for the special cases of low-loss media and good conductors. If media 1 and 2 have constitutive parameters $(\varepsilon_1, \mu_1, \sigma_1)$ and $(\varepsilon_2, \mu_2, \sigma_2)$ (**Fig. 8-7**), then expressions for the electric and magnetic fields in media 1 and 2 can be obtained from Eqs. (8.11) through (8.14) of **Table 8-1** by replacing jk with γ and η with η_c. Thus,

Medium 1

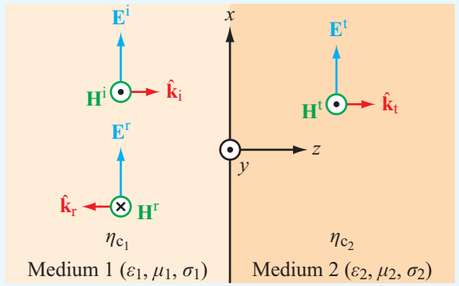
$$\widetilde{\mathbf{E}}_1(z) = \hat{\mathbf{x}} E^i_0 (e^{-\gamma_1 z} + \Gamma e^{\gamma_1 z}), \qquad (8.22a)$$

$$\widetilde{\mathbf{H}}_1(z) = \hat{\mathbf{y}} \frac{E^i_0}{\eta_{c_1}} (e^{-\gamma_1 z} - \Gamma e^{\gamma_1 z}), \qquad (8.22b)$$

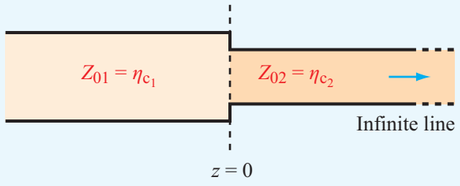
Medium 2

$$\widetilde{\mathbf{E}}_2(z) = \hat{\mathbf{x}} \tau E^i_0 e^{-\gamma_2 z}, \qquad (8.23a)$$

$$\widetilde{\mathbf{H}}_2(z) = \hat{\mathbf{y}} \tau \frac{E^i_0}{\eta_{c_2}} e^{-\gamma_2 z}. \qquad (8.23b)$$



(a) Boundary between dielectric media



(b) Transmission-line analogue

Figure 8-7 Normal incidence at a planar boundary between two lossy media.

Here, $\gamma_1 = \alpha_1 + j\beta_1$, $\gamma_2 = \alpha_2 + j\beta_2$, and

$$\Gamma = \frac{\eta_{c_2} - \eta_{c_1}}{\eta_{c_2} + \eta_{c_1}}, \qquad (8.24a)$$

$$\tau = 1 + \Gamma = \frac{2\eta_{c_2}}{\eta_{c_2} + \eta_{c_1}}. \qquad (8.24b)$$

Because η_{c_1} and η_{c_2} are, in general, complex, Γ and τ may be complex as well.

Example 8-3: Normal Incidence on a Metal Surface

A 1 GHz x-polarized plane wave traveling in the $+z$ direction is incident from air upon a copper surface. The air-to-copper interface is at $z = 0$, and copper has $\varepsilon_r = 1$, $\mu_r = 1$, and $\sigma = 5.8 \times 10^7$ S/m. If the amplitude of the electric field of the incident wave is 12 (mV/m), obtain expressions for the

因此，$|\Gamma| = 0.2$ 和 $\theta_r = \pi$。从方程。 (8.16) 电场强度最大时

$$l_{\max} = \frac{\theta_r \lambda_1}{4\pi} + n\frac{\lambda_1}{2} = \frac{\lambda_1}{4} + n\frac{\lambda_1}{2} \qquad (n = 0, 1, 2, \ldots)$$

与 $\lambda_1 = 0.6\ \mu$ 米。

(b)

$$S = \frac{1+|\Gamma|}{1-|\Gamma|} = \frac{1+0.2}{1-0.2} = 1.5.$$

(c) 传输到玻璃介质中的入射功率的比例等于方程（1）给出的传输功率密度之比。 (8.20) 到入射功率密度 $S^i_{av} = |E^i_0|^2/2\eta_1$：

$$\frac{S_{av_2}}{S^i_{av}} = \tau^2 \frac{|E^i_0|^2}{2\eta_2} \Big/ \left[\frac{|E^i_0|^2}{2\eta_1}\right] = \tau^2 \frac{\eta_1}{\eta_2}.$$

鉴于方程。 （8.21），

$$\frac{S_{av_2}}{S^i_{av}} = 1 - |\Gamma|^2 = 1 - (0.2)^2 = 0.96, \text{ or } 96\%.$$

8-1.4 有损媒体之间的边界

在第 8-1.1 节中，我们考虑了无损介质中的平面波垂直入射到另一个无损介质的平面边界上。我们现在将我们的表达推广到有损媒体。在具有本构参数（ε、μ、σ）的介质中，传播常数 $\gamma = \alpha + j\beta$ 和固有阻抗 η_c 都是复数。 α、β 和 η_c 的一般表达式由等式 1 给出。分别为（7.66a）、（7.66b）和（7.70），表7-2给出了低损耗介质和良导体特殊情况的近似表达式。如果介质1和介质2具有本构参数(ε_1、μ_1、σ_1)和(ε_2、μ_2、σ_2)(图8-7)，则介质1和介质2中的电场和磁场的表达式可以由式(1)获得。表 8-1 的（8.11）到（8.14），用 γ 替换 jk，用 η_c 替换 η。因此，

中1

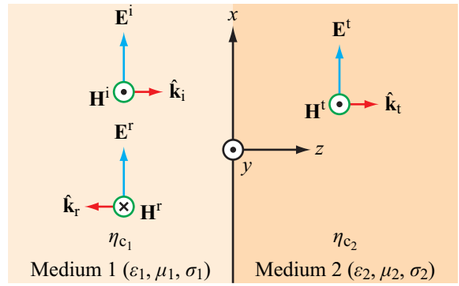
$$\widetilde{\mathbf{E}}_1(z) = \hat{\mathbf{x}} E^i_0 (e^{-\gamma_1 z} + \Gamma e^{\gamma_1 z}), \qquad (8.22a)$$

$$\widetilde{\mathbf{H}}_1(z) = \hat{\mathbf{y}} \frac{E^i_0}{\eta_{c_1}} (e^{-\gamma_1 z} - \Gamma e^{\gamma_1 z}), \qquad (8.22b)$$

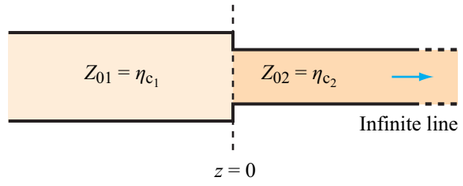
中2

$$\widetilde{\mathbf{E}}_2(z) = \hat{\mathbf{x}} \tau E^i_0 e^{-\gamma_2 z}, \qquad (8.23a)$$

$$\widetilde{\mathbf{H}}_2(z) = \hat{\mathbf{y}} \tau \frac{E^i_0}{\eta_{c_2}} e^{-\gamma_2 z}. \qquad (8.23b)$$



(a) Boundary between dielectric media



(b) Transmission-line analogue

图 8-7 两个有损介质之间平面边界处的法向入射。

这里，$\gamma_1 = \alpha_1 + j\beta_1$、$\gamma_2 = \alpha_2 + j\beta_2$ 和

$$\Gamma = \frac{\eta_{c_2} - \eta_{c_1}}{\eta_{c_2} + \eta_{c_1}}, \qquad (8.24a)$$

$$\tau = 1 + \Gamma = \frac{2\eta_{c_2}}{\eta_{c_2} + \eta_{c_1}}. \qquad (8.24b)$$

由于η_{c_1}和η_{c_2}通常是复杂的，因此Γ和τ也可能是复杂的。

示例 8-3：金属表面上的法向入射

沿 +z 方向行进的 1 GHz x 偏振平面波从空气入射到铜表面。空气-铜界面位于 z = 0，铜具有 $\varepsilon_r = 1$、$\mu_r = 1$ 和 $\sigma = 5.8 \times 10^7$ S/m。如果入射波的电场幅度为 12 (mV/m)，则得到表达式

instantaneous electric and magnetic fields in the air medium. Assume the metal surface to be several skin depths deep.

Solution: In medium 1 (air), $\alpha = 0$,

$$\beta = k_1 = \frac{\omega}{c} = \frac{2\pi \times 10^9}{3 \times 10^8} = \frac{20\pi}{3} \quad \text{(rad/m)},$$

$$\eta_1 = \eta_0 = 377\ (\Omega), \qquad \lambda = \frac{2\pi}{k_1} = 0.3\ \text{m}.$$

At $f = 1$ GHz, copper is an excellent conductor because

$$\frac{\varepsilon''}{\varepsilon'} = \frac{\sigma}{\omega\varepsilon_r\varepsilon_0} = \frac{5.8 \times 10^7}{2\pi \times 10^9 \times (10^{-9}/36\pi)} = 1 \times 10^9 \gg 1.$$

Use of Eq. (7.77c) gives

$$\eta_{c_2} = (1+j)\sqrt{\frac{\pi f \mu}{\sigma}} = (1+j)\left[\frac{\pi \times 10^9 \times 4\pi \times 10^{-7}}{5.8 \times 10^7}\right]^{1/2}$$
$$= 8.25(1+j) \quad (\text{m}\Omega).$$

Since η_{c_2} is so small compared to $\eta_0 = 377\ (\Omega)$ for air, the copper surface acts, in effect, like a short circuit. Hence,

$$\Gamma = \frac{\eta_{c_2} - \eta_0}{\eta_{c_2} + \eta_0} \approx -1.$$

Upon setting $\Gamma = -1$ in Eqs. (8.11) and (8.12) of **Table 8-1**, we obtain

$$\widetilde{\mathbf{E}}_1(z) = \hat{\mathbf{x}} E_0^{\text{i}}(e^{-jk_1 z} - e^{jk_1 z}) = -\hat{\mathbf{x}} j 2E_0^{\text{i}} \sin k_1 z, \qquad (8.25a)$$

$$\widetilde{\mathbf{H}}_1(z) = \hat{\mathbf{y}} \frac{E_0^{\text{i}}}{\eta_1}(e^{-jk_1 z} + e^{jk_1 z}) = \hat{\mathbf{y}} 2 \frac{E_0^{\text{i}}}{\eta_1} \cos k_1 z. \qquad (8.25b)$$

With $E_0^{\text{i}} = 12$ (mV/m), the instantaneous fields associated with these phasors are

$$\mathbf{E}_1(z,t) = \Re\mathrm{e}[\widetilde{\mathbf{E}}_1(z)\, e^{j\omega t}]$$
$$= \hat{\mathbf{x}} 2E_0^{\text{i}} \sin k_1 z \sin \omega t$$
$$= \hat{\mathbf{x}} 24 \sin(20\pi z/3) \sin(2\pi \times 10^9 t) \quad (\text{mV/m}),$$
$$\mathbf{H}_1(z,t) = \Re\mathrm{e}[\widetilde{\mathbf{H}}_1(z)\, e^{j\omega t}]$$
$$= \hat{\mathbf{y}} 2 \frac{E_0^{\text{i}}}{\eta_1} \cos k_1 z \cos \omega t$$
$$= \hat{\mathbf{y}} 64 \cos(20\pi z/3) \cos(2\pi \times 10^9 t) \quad (\mu\text{A/m}).$$

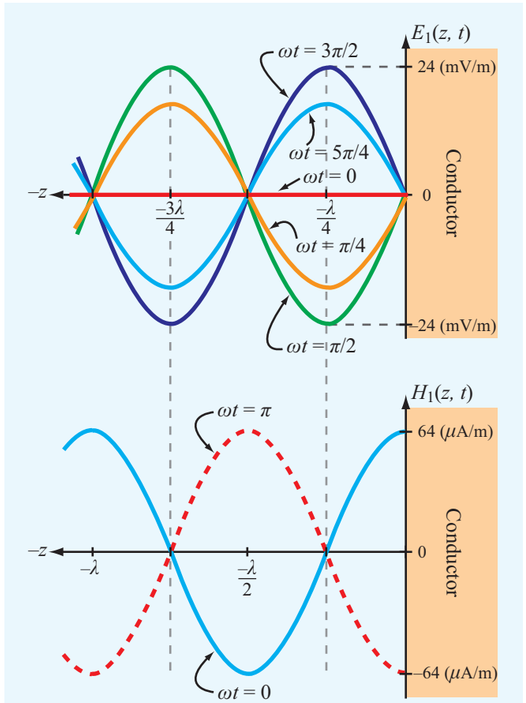


Figure 8-8 Wave patterns for fields $E_1(z,t)$ and $H_1(z,t)$ of Example 8-3.

Plots of the magnitude of $\mathbf{E}_1(z,t)$ and $\mathbf{H}_1(z,t)$ are shown in **Fig. 8-8** as a function of negative z for various values of ωt. The wave patterns exhibit a repetition period of $\lambda/2$, and E and H are in phase quadrature (90° phase shift) in both space and time. This behavior is identical with that for voltage and current waves on a shorted transmission line.

Concept Question 8-1: What boundary conditions were used in the derivations of the expressions for Γ and τ?

Concept Question 8-2: In the radar radome design of Example 8-1, all the incident energy in medium 1 ends up getting transmitted into medium 3, and vice versa. Does this imply that no reflections take place within medium 2? Explain.

空气介质中的瞬时电场和磁场。假设金属表面有几个趋肤深度。

解：在介质1（空气）中，$\alpha = 0$，

$$\beta = k_1 = \frac{\omega}{c} = \frac{2\pi \times 10^9}{3 \times 10^8} = \frac{20\pi}{3} \quad \text{(rad/m)},$$

$$\eta_1 = \eta_0 = 377\ (\Omega), \qquad \lambda = \frac{2\pi}{k_1} = 0.3\ \text{m}.$$

在 f = 1 GHz 时，铜是一种优良的导体，因为

$$\frac{\varepsilon''}{\varepsilon'} = \frac{\sigma}{\omega\varepsilon_r\varepsilon_0} = \frac{5.8 \times 10^7}{2\pi \times 10^9 \times (10^{-9}/36\pi)} = 1 \times 10^9 \gg 1.$$

使用等式。(7.77c) 给出

$$\eta_{c_2} = (1+j)\sqrt{\frac{\pi f \mu}{\sigma}} = (1+j)\left[\frac{\pi \times 10^9 \times 4\pi \times 10^{-7}}{5.8 \times 10^7}\right]^{1/2}$$
$$= 8.25(1+j) \quad (\text{m}\Omega).$$

由于 η_{c_2} 与空气中的 $\eta_0 = 377\ (\Omega)$ 相比非常小，因此铜表面实际上就像短路一样。因此，

$$\Gamma = \frac{\eta_{c_2} - \eta_0}{\eta_{c_2} + \eta_0} \approx -1.$$

在等式中设置 $\Gamma = -1$ 后。由表8-1的(8.11)和(8.12)，我们得到

$$\widetilde{\mathbf{E}}_1(z) = \hat{\mathbf{x}} E_0^{\text{i}}(e^{-jk_1 z} - e^{jk_1 z}) = -\hat{\mathbf{x}} j 2E_0^{\text{i}} \sin k_1 z, \qquad (8.25a)$$

$$\widetilde{\mathbf{H}}_1(z) = \hat{\mathbf{y}} \frac{E_0^{\text{i}}}{\eta_1}(e^{-jk_1 z} + e^{jk_1 z}) = \hat{\mathbf{y}} 2 \frac{E_0^{\text{i}}}{\eta_1} \cos k_1 z. \qquad (8.25b)$$

对于 $E_0^{\text{i}} = 12$ (mV/m)，与这些相量相关的瞬时场为

$$\mathbf{E}_1(z,t) = \Re\mathrm{e}[\widetilde{\mathbf{E}}_1(z)\, e^{j\omega t}]$$
$$= \hat{\mathbf{x}} 2E_0^{\text{i}} \sin k_1 z \sin \omega t$$
$$= \hat{\mathbf{x}} 24 \sin(20\pi z/3) \sin(2\pi \times 10^9 t) \quad (\text{mV/m}),$$
$$\mathbf{H}_1(z,t) = \Re\mathrm{e}[\widetilde{\mathbf{H}}_1(z)\, e^{j\omega t}]$$
$$= \hat{\mathbf{y}} 2 \frac{E_0^{\text{i}}}{\eta_1} \cos k_1 z \cos \omega t$$
$$= \hat{\mathbf{y}} 64 \cos(20\pi z/3) \cos(2\pi \times 10^9 t) \quad (\mu\text{A/m}).$$

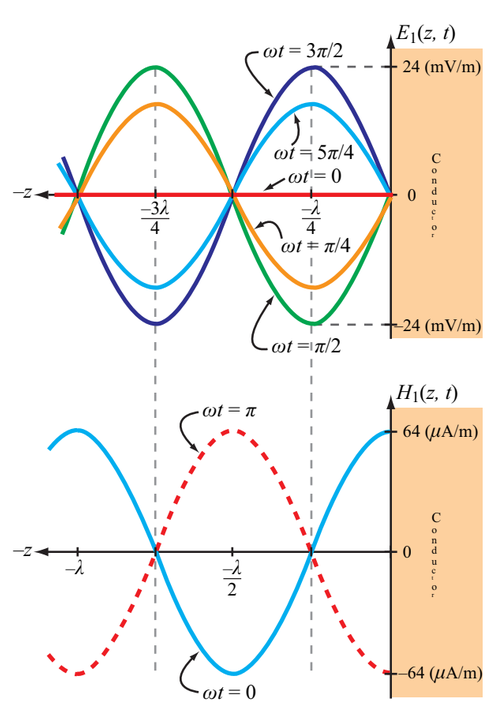


图 8-8 例 8-3 中 $E_1(z,t)$ 和 $H_1(z,t)$ 场的波形。

$E_1(z,t)$ 和 $H_1(z,t)$ 的幅值图如图 8-8 所示，作为不同 ωt 值的负 z 的函数。波形的重复周期为$\lambda/2$，E 和H 在空间和时间上相位正交（90° 相移）。这种行为与短路传输线上的电压和电流波的行为相同。

概念问题 8-1：在 Γ 和 τ 表达式的推导中使用了哪些边界条件？

概念问题 8-2：在例 8-1 的雷达天线罩设计中，介质 1 中的所有入射能量最终都会传输到介质 3 中，反之亦然。这是否意味着介质 2 内不会发生反射？解释。